

# OneRoot AgriRadius Pro

## New-user operating guide

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Turn any place in South India into a complete farming report from free satellite and government data — crops, soil, rainfall, live weather, market prices and livestock/allied sectors.

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### 1. Opening

- Use **Google Chrome** (desktop or phone). No installation.
- If asked, type the **password** you were given.
- **First load can take ~30 s** if the app was asleep — then it's fast.

### 2. The screen

- **Left sidebar** = all your controls (Search, Input, Layers, Compute quality, Service health).
- **Main area** = the map on top, and **Analysis Results** tabs below.
- On a phone the sidebar folds into the menu (top-left).

### 3. Choose what to analyse (sidebar Input)

- **Analysis Mode:** *Area (radius)* studies a circle; *Point location* studies one exact spot.
- **Input Method:** type coordinates, or pick **Map Click** and click the map. You can also paste a **Google Maps link** or place name in the Search box at the top.
- **Radius (km):** slider or exact value — keep it **7–10 km** (big circles are slow and use more compute).
- Pick the **Year** for the satellite analysis. The **current year** uses a rolling 12-month window so detectors still work mid-season; a completed past year is the safest reference.
- New here? Click **Try a sample area (Pollachi)** in the sidebar to load a demo location instantly.
- *Example:* 10 . 6588, 77 . 0089 (Pollachi), radius 10 km.

### 4. The map (sidebar Layers)

- Tick a layer to turn it on — it computes live the first time (10–30 s), then loads instantly.
- **Overlay opacity** makes layers see-through.
- **Compute quality** (Light / Balanced / Heavy): Heavy = sharpest but heaviest — drop to **Light** if it's slow or Earth Engine is busy.
- If tiles look missing after zooming, click **Refresh map**.
- **Compare layers (grid):** above the map, switch *Map view* to *Compare layers (grid)* to see every ticked layer in its own panel side by side. Toggle **synced zoom** (all panels follow the top-left master — pan/zoom it and the rest match) or make them independent, set **max panels**, and use **Full screen**. Each panel shows its

key figure on a translucent badge.

Layers include: Dynamic World land cover, Cropland Confidence, Paddy, Maize, Plantations (coconut/arecanut), Banana, Aquaculture ponds, Soil pH / Organic Carbon / Nitrogen.

## 5. The results tabs (below the map)

Some tabs fetch data only when you click a button, so nothing loads unless you ask.

- **Summary** — land-cover mix, cropland confidence & 3-year stability.
- **Villages** — per-village cropland & ranking; the trained crop classifier (incl. the coconut model).
- **Charts** — visual breakdown.
- **Crop Cycle** — sowingpeakharvest pattern; paddy & plantation checks.
- **Rainfall** — 10-year history.
- **Forecast** — **Live conditions now** (rain, temperature, humidity, wind, sun/solar, UV, soil moisture & temp, evapotranspiration, and a **drying-suitability score**) with an *Auto 5m* refresh toggle, plus the 16-day outlook & dry-window.
- **Soil** — pH, organic carbon, nitrogen, texture, plus a **Land Capability** panel (SLUSI detailed soil survey): how much of the district's *surveyed* land is prime/arable (Class I-IV) vs non-arable (V-VIII), with links to the official survey PDFs. It's authoritative but **historical reference** (surveys 1960-2018, surveyed watersheds only) — not current land use.
- **Allied Sectors** — livestock & poultry, estimated **dairy pool & feed demand**, aquaculture, sericulture, fisheries, fertiliser, horticulture.
- **Mandi** — today's prices, the **MSP floor** comparison, a multi-year **price trend**, and a **variety/grade** breakdown.
- **Ground Truth** — log what a field grows (trains the app) + soil cards. The **Upload Field Data** tab lets you collect points two ways: **on-site on your phone** (tap *Get my location*, pick the crop — or *Not sure* to skip — and Save, one tap per field), or **upload a CSV/Excel** of many points at once (lat, lon, crop; village/acreage/notes optional; template provided). Everything saves to the shared dataset for calibration and the classifier.
- **Downloads** — build a full **PDF / Excel report** of everything.

## 6. How much to trust it

Open the **Data & Confidence** box at the top of the results.

**Measured** things (rainfall, crop vigour, radar detection, prices) are reliable; **modelled/classified** things (land-cover class, soil at 250 m) are best read as **ranges**. Rule of thumb: \*trust the direction and the ranges; verify the edges on the ground.\* It improves as your team logs ground truth.

**Know how current each number is.** Every tab shows a small dated caption ( live · periodic release · modelled · historical reference), and the **Data & Confidence** box lists an "as of" date for every source. Live weather and mandi prices are today's; the livestock census is 2019; soil (SoilGrids) is a 2020 modelled baseline; the SLUSI land-capability survey is 1960-2018. Always read the date so you don't mistake older reference data for today's ground reality.

## 7. Please test mindfully

This is a **free, open-source** setup (Google Earth Engine) with a **limited shared monthly compute budget**:

- one heavy layer at a time; keep the radius reasonable;
- avoid rapid repeated clicks — let a layer finish;
- use **Light** compute quality if slow.

The sidebar **Service health** panel shows the live EECU compute meter (how much is left; it resets on the 1st of each month).

## Quick fixes

- *Slow first open?* It was asleep — wait ~30 s.
- *"Earth Engine is busy"?* Wait a minute, click Refresh, or use Light.
- *Tiles missing after zoom?* Click **Refresh map**.
- *Map jumped?* Re-search your location.

## How current is each data source?

live · periodic government release · modelled baseline · historical reference (read on 2026-07-27)

| Data source                                       | As of                                        | Type                 |
|---------------------------------------------------|----------------------------------------------|----------------------|
| Sentinel-2 optical + Dynamic World land cover     | selected analysis year (season composite)    | periodic release     |
| SoilGrids v2.0 (ISRIC)                            | 2020 release · modelled, 250 m               | modelled baseline    |
| ERA5-Land soil temperature & moisture             | monthly, ~1-2 month lag                      | periodic release     |
| CHIRPS rainfall history                           | 1981 ~1 month ago                            | periodic release     |
| Open-Meteo current & forecast                     | live (updated hourly)                        | live                 |
| 20th Livestock Census                             | 2019                                         | periodic release     |
| AGMARKNET APMC mandi prices                       | daily (latest reported arrivals)             | live                 |
| Minimum Support Price (CACP / GoI)                | 2025-26 season                               | periodic release     |
| Horticulture area & production (State Hort. Dept) | ~2022-23                                     | periodic release     |
| Fertiliser consumption (DES)                      | 2022-23                                      | periodic release     |
| Land-use statistics (DES)                         | ~2020-22                                     | periodic release     |
| ESA WorldCereal cropland                          | 2021                                         | periodic release     |
| ESA WorldCover v200                               | 2021                                         | periodic release     |
| SLUSI Detailed Soil Survey — land capability      | field surveys 1960-2018 (varies by district) | historical reference |

Always read the 'As of' date so you never mistake older reference data (e.g. the 1960-2018 SLUSI soil survey or the 2019 livestock census) for today's ground reality.